Enterprise VLSI Chip Design & Verification System

Project Report

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Project: 32-bit Single-Cycle Processor using Verilog HDL

Abstract

This project presents the design and verification of a 32-bit single-cycle processor using Verilog HDL. The design includes a Program Counter, Instruction Memory, Register File, ALU, Control Unit, Data Memory, and a top-level processor. Functional verification is performed using ModelSim and synthesis is targeted with Quartus Prime Lite.

Objectives

Design RTL, verify functionality, synthesize the processor, analyze timing/resources, and document the complete workflow.

Architecture

Processor flow: Program Counter → Instruction Memory → Control Unit → Register File → ALU → Data Memory → Write Back.

Modules

Program Counter, Instruction Memory, Register File, ALU, Control Unit, Data Memory, Top Processor.

Instruction Set

MOVI, ADD, SUB, AND, OR, LOAD, STORE, NOP.

Verification

Each module was verified using dedicated ModelSim testbenches. The complete processor was verified using processor_tb.

Synthesis

Compile the design in Quartus Prime Lite and attach RTL Viewer, Technology Map Viewer, Compilation Report, Timing Report, and resource utilization screenshots.

DFT

The design uses modular verification. Future improvements include scan-chain and boundary-scan support.

Conclusion

The processor demonstrates a complete RTL design and verification workflow suitable for an academic/internship project.